



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO VELAVAN FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

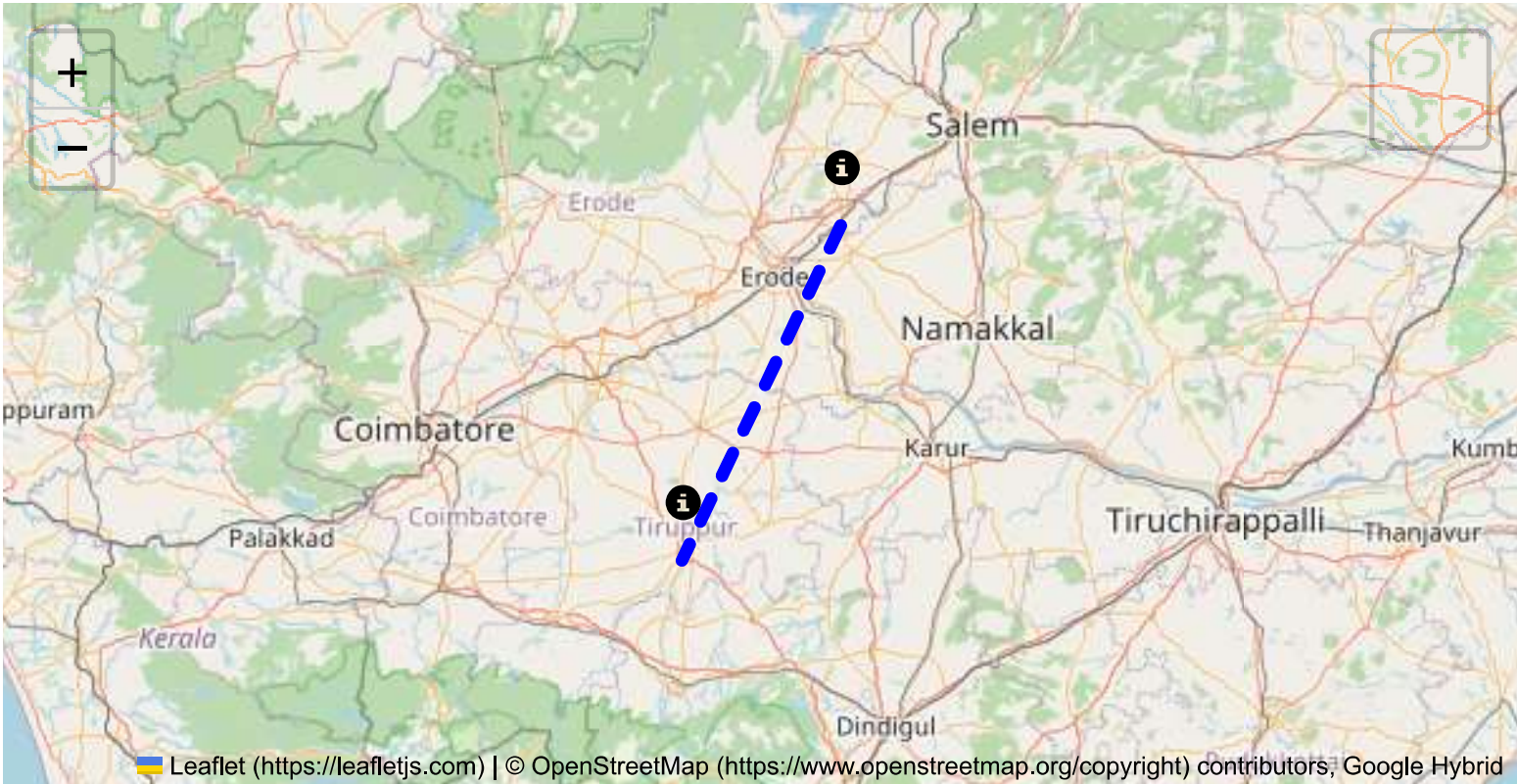
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

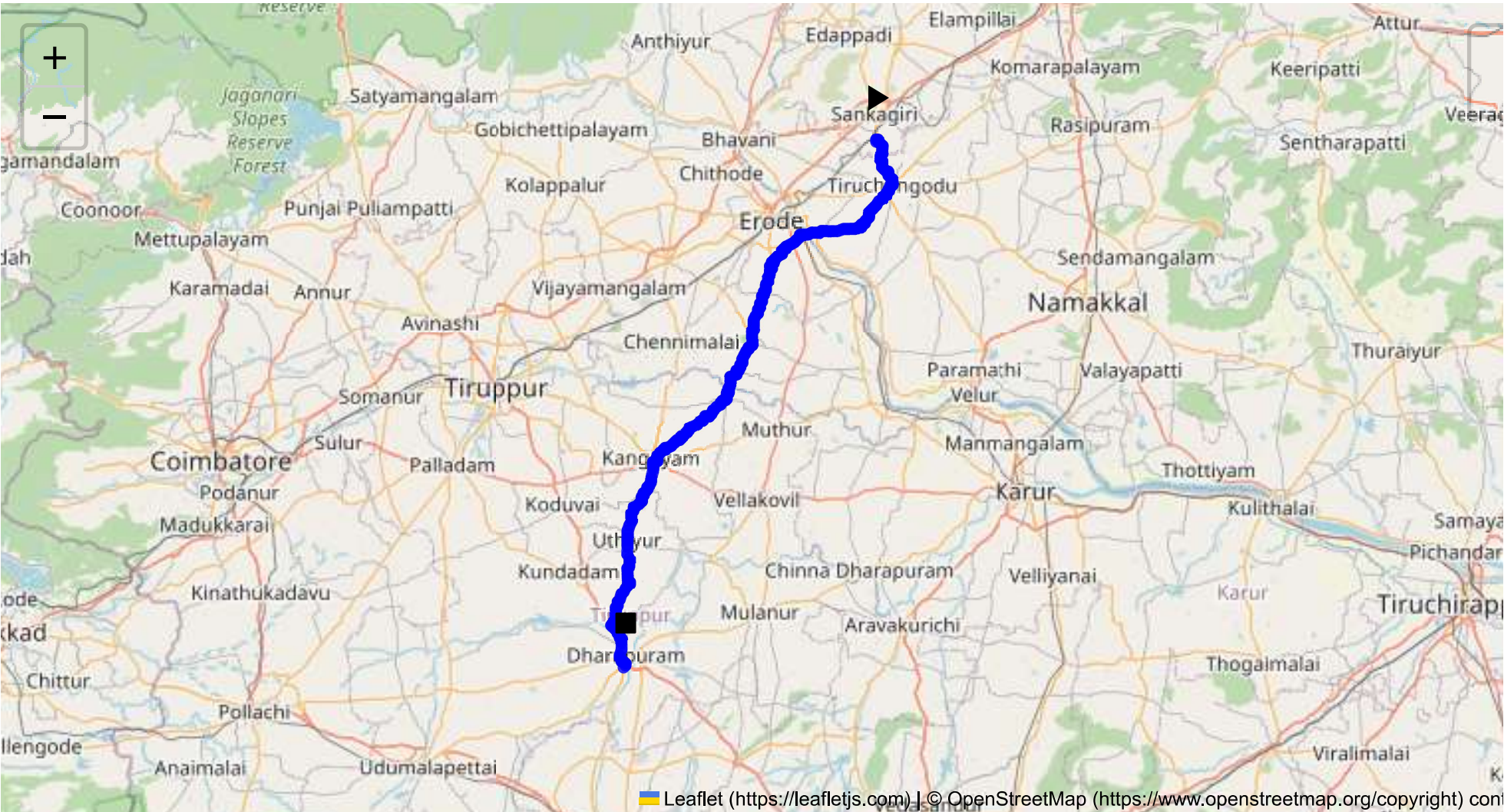
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 104.61 km**  
**Estimated Duration: 2.4 hours**  
**Adjusted Duration (Heavy Vehicle): 3.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.719177, 77.522675)**



**Welcome to the Journey Risk Management Study**

**Route Safety Analysis: Sangagiri to Kurunji Nagar, Tamil Nadu**

**1. Overview of the Route Map:** The route from Sangagiri to Kurunji Nagar stretches approximately 104.61 kilometers, primarily via Tamil Nadu State Highway 21. The path passes through rural areas and small towns, providing a mix of straight stretches and winding segments typical of Indian state highways.



**2. Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The monsoon season (June to September) brings heavy rainfall, potentially leading to flooded roads and reduced visibility. During the dry season (November to February), fog may occur in the early morning, presenting visibility challenges. The summer months (March to May) bring extreme heat, which can affect vehicle performance.

**3. Analysis of Traffic Patterns:** Traffic is generally moderate, but peak hours occur from 8:00 AM to 10:00 AM and 4:00 PM to 7:00 PM, especially near towns like Tiruchengode. Major market days and festivals can increase congestion. The main congestion-prone areas are urban centers, particularly near intersections and junctions.

**4. Assessment of Road Quality and Infrastructure:** The road quality varies, with some well-paved sections and others requiring maintenance. Potholes and uneven surfaces may be present, particularly after monsoons. Roadside infrastructure, including signage and lighting, can be sporadic in rural segments.

**5. Suggestions for Alternative Routes for Emergencies:** An alternative route is via NH 544, although it increases the travel distance. This national highway offers better road conditions and connectivity to emergency services.

**6. Summary of Local Regulations Affecting Hazardous Material Transport:** Transporting hazardous materials in Tamil Nadu requires adherence to national guidelines, including specific licenses and vehicle markings. Routing through residential and dense commercial areas should be minimized.

**7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** While specific historical incidents along this route are limited, heavy vehicles have occasionally faced issues with road conditions leading to accidents. Incidents involving hazardous materials usually involve inadequate containment or handling practices.

**8. Environmental Considerations and Sensitive Areas:** The route crosses several agricultural zones, necessitating care to prevent contamination. Drinking water sources close to Dharapuram are sensitive to pollutants.

**9. Analysis of Communication Coverage:** Mobile network coverage is generally reliable, but rural stretches may have brief dead zones. Providers like Jio and Airtel cover most areas, but drivers should be prepared for occasional disruptions.

**10. Estimated Emergency Response Times for Different Route Segments:** Urban areas like Tiruchengode have quicker emergency response times (about 15-20 minutes), while rural segments could see responses extended to 30-45 minutes or more.

**12. Overall Summary of Risk Assessment:** The route entails moderate risk, primarily due to variable road conditions and weather-related hazards. Traffic congestion is generally manageable, but specific legislative compliance is necessary for hazardous material transport. Drivers should remain vigilant, especially during the monsoon season, and maintain regular communication with logistics and emergency services. Coordination for quicker response and adherence to safety and environmental protocols is crucial for mitigating potential risks.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Roundabout | High       | 11.00172, 77.56268 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44022, 77.87546 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.39976, 77.88382 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.39690, 77.88852 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.36616, 77.88885 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.34067, 77.86347 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.33854, 77.86216 | 15 KM/Hr    |
| 9  | Turn       | Medium     | 11.33778, 77.86236 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.30952, 77.77587 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.30973, 77.76712 | 30 KM/Hr    |
| 12 | Turn       | High       | 11.29490, 77.74685 | 15 KM/Hr    |
| 13 | Turn       | High       | 11.29125, 77.74646 | 15 KM/Hr    |
| 14 | Turn       | Medium     | 11.28844, 77.74319 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.28573, 77.74281 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.27319, 77.72981 | 15 KM/Hr    |
| 17 | Blind Spot | Blind Spot | 11.27330, 77.72715 | 10 KM/Hr    |
| 18 | Turn       | Medium     | 11.23054, 77.71796 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.20776, 77.70916 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.15942, 77.70068 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.12102, 77.67919 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.11932, 77.67786 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.11710, 77.67175 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.10436, 77.67044 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.09491, 77.66603 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.08696, 77.66407 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.08296, 77.65857 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.06183, 77.63924 | 30 KM/Hr    |
| 29 | Turn       | High       | 11.00826, 77.56962 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 30 | Blind Spot | Blind Spot | 11.00058, 77.56882 | 10 KM/Hr    |
| 31 | Turn       | High       | 10.94722, 77.54709 | 15 KM/Hr    |
| 32 | Turn       | Medium     | 10.93895, 77.53976 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 10.93054, 77.53265 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 10.89213, 77.52646 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 10.83409, 77.52932 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 10.77642, 77.50712 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 10.77580, 77.50724 | 30 KM/Hr    |
| 38 | Turn       | Medium     | 10.75864, 77.51918 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 10.75722, 77.51935 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 10.73670, 77.51731 | 30 KM/Hr    |
| 41 | Blind Spot | Blind Spot | 10.72415, 77.52567 | 10 KM/Hr    |

## Emergency Locations

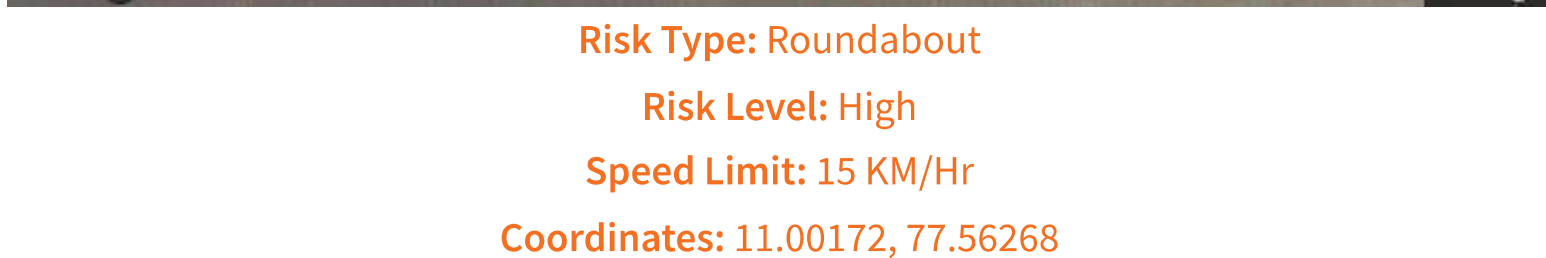
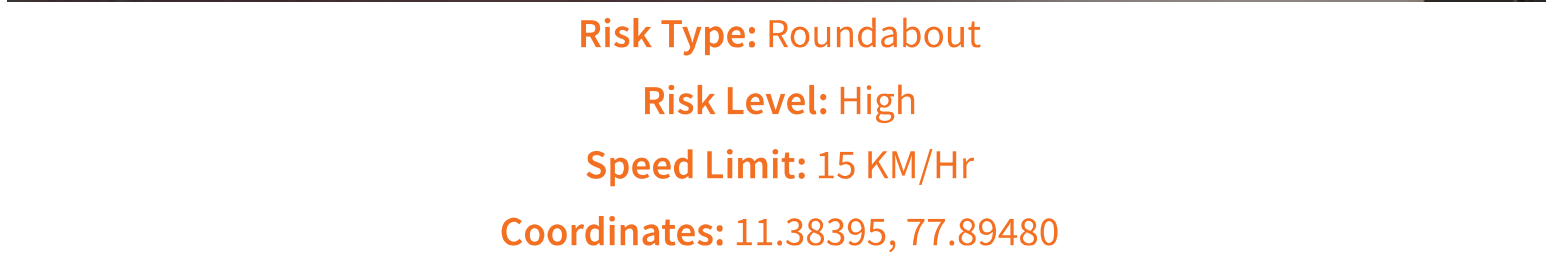
|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 6  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 10 | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 13 | hospital | Government Hospital              | 11.2343326, 77.719226  | 30 km/h     | Medium     |
| 16 | police   | Kangeyam Police Station          | 11.0011411, 77.562688  | 30 km/h     | Medium     |
| 17 | hospital | J K Hospital                     | 11.0042298, 77.5603345 | 30 km/h     | Medium     |
| 21 | hospital | Sri Balaji Surya Hospital        | 10.735518, 77.5185388  | 30 km/h     | Medium     |
| 22 | hospital | Ramana Hospital, Dharapuram      | 10.7351092, 77.517107  | 30 km/h     | Medium     |
| 24 | hospital | Janani Hospital, Dharapuram      | 10.7294033, 77.5210119 | 30 km/h     | Medium     |
| 25 | clinic   | Punitha Nursing Home             | 10.72901, 77.520148    | 30 km/h     | Medium     |
| 26 | hospital | Ponnaiah Hospital, Dharapuram    | 10.72916, 77.520131    | 30 km/h     | Medium     |

Crowded Spots

|    | type        | name                                                 | coordinates               | speed_limit | risk_level |
|----|-------------|------------------------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                         | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                        | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை                  | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution                          | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                                           | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |
| 14 | college     | Government Arts and Science College, Kangeyam        | 11.0626738,<br>77.636227  | 30 km/h     | Medium     |
| 15 | school      | Carmel Girls Higher Secondary School                 | 11.0009019,<br>77.5710182 | 30 km/h     | Medium     |
| 18 | school      | Global Matriculation Higher Secondary School         | 10.965994,<br>77.5591898  | 30 km/h     | Medium     |
| 19 | college     | Maharani College                                     | 10.7625655,<br>77.5157126 | 30 km/h     | Medium     |
| 20 | school      | Vishnu Lakshmi Matriculation Higher Secondary School | 10.7340392,<br>77.5164506 | 30 km/h     | Medium     |
| 23 | school      | Thenmalar Higher Secondary School                    | 10.7333748,<br>77.5166481 | 30 km/h     | Medium     |

Route Photos of Risky Spots







**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44022, 77.87546



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39976, 77.88382



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39690, 77.88852





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.36616, 77.88885



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.34067, 77.86347





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.33854, 77.86216**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.33778, 77.86236**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.30952, 77.77587**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.30973, 77.76712



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29490, 77.74685





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29125, 77.74646



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28844, 77.74319



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.28573, 77.74281





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.27319, 77.72981



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.27330, 77.72715





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.23054, 77.71796



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.20776, 77.70916



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.15942, 77.70068





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.12102, 77.67919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.11932, 77.67786





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.11710, 77.67175



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.10436, 77.67044



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.09491, 77.66603





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08696, 77.66407



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08296, 77.65857





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.06183, 77.63924



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.00826, 77.56962



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.00058, 77.56882





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.94722, 77.54709



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.93895, 77.53976





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.93054, 77.53265



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.89213, 77.52646



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.83409, 77.52932





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.77642, 77.50712



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.77580, 77.50724





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.75864, 77.51918



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.75722, 77.51935



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.73670, 77.51731





**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 10.72415, 77.52567